Toward an Advanced Method for Full-Waveform Hyperspectral LiDAR Data Processing

Jie Bai, Zheng Niu, Kaiyi Bi, Xuebo Yang, Yanru Huang, Yuwen Fu, Mingquan Wu, and Li Wang, *Member, IEEE*

***Abstract*— Full-waveform hyperspectral LiDAR (HSL) generates comprehensive hyperspectral waveforms for scenes to reveal the shape and spectral heterogeneity of multiple natural targets. Nevertheless, current waveform processing methods are primarily designed for single-wavelength light detection and ranging (LiDAR) systems, resulting in a shortage of methods tailored for full-waveform HSL data processing and in a restriction to further quantitative applications for HSL. This study is designed to extract targets' physical and spectral characteristics by integrating spectral-dimension features into the HSL waveform processing. The core idea of the method involves a rigorous processing technique consisting of parameter initialization, parameter optimization, and reoptimization over calculating the median (M) after ranking central locations of natural target echoes (Rclonte). The medians in the reoptimization step serve as the reference parameter sets for supplementing the hidden or weak components at some wavelengths for HSL. Two groups of datasets, the simulated and measured datasets, were utilized to evaluate the component detection ability of the proposed Rclonte-M method. The results suggest that the Rclonte-M method demonstrates excellent component detection performance on both simulated and measured data, outperforming the multispectral waveform decomposition (MSWD) method. The HSL system designed by us owns an overall ranging error of about 7 cm for adjacent components, with the relative neighbor distance error (RNDE) limited to 0.160. Besides, spectra retrieval results from HSL easily distinguish the natural targets along the laser path. This study enriches the full-waveform HSL data processing algorithm library and could be considered in other full-waveform HSL systems and the simulated airborne or spaceborne HSL waveforms. Codes are freely available on https://github.com/Jie-Bai/Rclonte-M-TGRS.**

***Index Terms*— Full-width at half-maximum (FWHM), Gaussian decomposition, multiple natural targets, range resolution, ranking central locations of natural target echoes (Rclonte)-M, waveform processing.**

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I. Introduction

LIGHT detection and ranging (LiDAR) is an active remote sensing tool owning the penetration feature, which is adept at presenting point clouds of targets with the geometric locations and corresponding intensity values [1], [2], [3], [4], [5]. In contrast to conventional discrete-waveform LiDAR by capturing the final discrete intensity values after interaction between the emitted laser beam and the targets along the laser path [6], [7], [8], [9], the full-waveform LiDAR data, in particular, record the detailed and complete signals containing several echoes after the complex interaction process [10], [11], [12], [13], [14], [15]. Thereby, the further handling on the client side for these backward full-waveform echoes enables the extraction of physical and geometric attributes of natural targets situated along the laser path [16], [17], [18]. Parameters, such as intensity, central location, full-width at half-maximum (FWHM), or some other representative parameters, such as cross section [19], [20], [21], [22], [23], [24], provide invaluable insights for target interpretation in the handling process. Generally, the penetration capability of full-waveform LiDAR offers an advantageous vertical perspective for target observation, unlocking enhanced potential in quantitative applications. Specifically, the characteristics derived from full-waveform data have found widespread utility across diverse fields, such as above-ground biomass (AGB) estimation [25], canopy height retrieval [26], tree classification and segmentation [24], [27], [28], [29], leaf chlorophyll content estimation [30], [31], ice thickness calculation [32], bathymetric surveying [33], [34], as well as other quantitative applications [14], [35].

Traditional waveform processing techniques have been predominantly tailored for single-wavelength LiDAR systems [2], [36], [37], [38], [39]. Due to spectrum-range constraints, namely, the limited wavelength counts, the single-wavelength decomposition results typically fall short of capturing the spectral profile of targets along the laser path [23], [40]. This limitation impedes a comprehensive characterization of target reflection properties. Meanwhile, in the component detection process, some hidden weak or overlapping waveforms generated by poorly reflecting or close neighboring targets could not be detected for the single-wavelength LiDAR, leading to incomplete component identification [41]. However, full-waveform hyperspectral LiDAR (HSL), with abundant wavelengths [42], [43], [44], [45], [46], [47], is regarded as a viable solution to the above issues and holds promise in obtaining a better component detection performance. Indeed,

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the HSL prototypes record the return hyperspectral waveforms, and thus, the components that might be missed at a specific wavelength could be potentially and successfully complemented and identified by other wavelengths.

Nonetheless, there are no abundant waveform processing techniques for the novel HSL systems untill now. The complex waveform shapes and inconsistent time delay effects within the hyperspectral waveforms present challenges in hyperspectral waveform processing. Traditional waveform processing methods for single-wavelength LiDAR may not be directly compatible. Previously reported studies in the literature pointed out that it was imperative to develop much more decomposition techniques tailored for HSL systems [40], [41], [48], [49], where a multispectral waveform decomposition (MSWD) method took the lead in decomposing multispectral LiDAR return waveforms and is widely regarded as a representative technique [40]. Its core principle revolves around energy-based descending sorting for scenes containing multiple diffuse-reflection targets. In such scenes, Gaussian components sharing the same energy level are conventionally assumed to correspond to the same target with high probability. Then, the average values were calculated to complement the hidden components. This approach is apt for diffuse-reflection or isotropic targets, such as standard reflectance panels. However, in scenarios encompassing multiple natural targets, waveform components with the same energy order across different wavelengths are more likely to correspond to distinct targets. Therefore, MSWD may not be compatible with the decomposition task where the scene was composed of multiple natural targets, instead of the Lambertian targets. In brief, in the context of appearing of the novel HSL waveform data, there arises an actual need for more novel and stable waveform decomposition techniques that can be utilized to process the complex return waveforms from the scene comprising multiple natural targets.

In this study, we introduced a novel HSL waveform decomposition method. It aims to make full use of the rich wavelength information of HSL in compensating for the missed or false detection of hidden weak and overlapping components. We carried out relevant experiments on an HSL instrument designed independently by our laboratory. The main objectives are developing a novel and advanced hyperspectral waveform processing technique to decompose the HSL waveform data, examining the ranging accuracy of adjacent components within a footprint for the HSL system designed by us, and exploring the retrieval potential of reflection property for the natural targets along the laser path using the HSL system.

II. Materials and Methods

A. HSL Instrument

The HSL system used in this work was independently designed and developed by our laboratory, as shown in Fig. 1. It can collect the full-waveform echo signal of the targets along the laser track within the spot at listed wavelengths in Fig. 1(a). The instrument mainly consists of a laser transmitting system, a receiving system, a scanning system, a spectroscopic system,

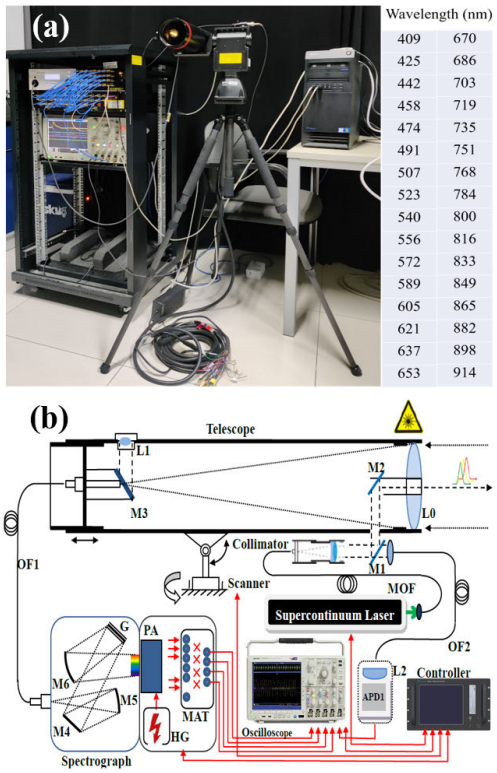


Fig. 1. Hyperspectral LiDAR system with a supercontinuum pulsed laser source. (a) Prototype image. (b) Internal optical structures. L0, L1, and L2 represent the achromatic lens. M1 is the microlens, while M2 and M4 stand for the reflectors. M3 refers to the diffuse mirror. M5 represents the collimation mirror. M6 is the converging mirror. G represents the grating spectrometer. APD1 stands for an avalanche photodiode. MAT denotes the switch matrix. OF1 and OF2 are optical fibers. MOF is the microstructure optical fiber. PA is a photosensitive array. HG represents the generator.

and a storage system [42], [43]. The laser transmitting system emits a beam of Gaussian white light of 400–2500 nm, with a pulse frequency of 20–40 kHz and a peak power of 20 kW. After interacting with the target, the emitted laser pulse returns to the receiving aperture. Then, it is split by a spectroscopic system (a grating) and converted into the electrical signal by a photodetector. Finally, the signal is converted into a digital signal through analog-to-digital conversion and displayed on an oscilloscope screen. The storage system records the displayed curves. The wavelength range of the HSL system covers from 409 to 914 nm with spectral resolution of about 16–17 nm (32 wavelengths in total as designed). Recent studies indicate that the middle-part wavelengths of our system usually own higher quality with low noise [42], [50]. The FWHM of the emitted pulse is about 1.7 ns. The laser divergence angle is 5 mrad, which means that the footprint diameter is 5 cm at a distance of 10 m. Detailed system configuration parameters can be found in the previously published study [43].

B. Simulated Data and Measured Data

For the simulated dataset, we crafted a scenario featuring four distinct types of natural targets: green leaf, ore, grassland, and sand. The spectral characteristics of these natural targets (see Fig. 2) were sourced from the USGS

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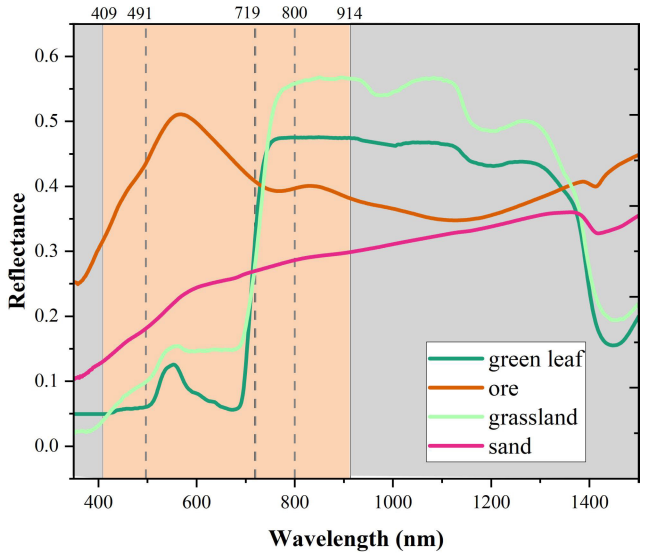


Fig. 2. Reflectance spectra of four natural targets from the USGS spectral library.

spectral library (available on the website: https://crustal.usgs.gov/speclab/QueryAll07a.php, last accessed on November 11, 2023). HSL intensities at various wavelengths were computed based on their spectrum ratios, with the initial intensity value of a green leaf at 409 nm set to 3 mV. A noise error, mirroring that of the HSL system, was introduced into the echo waveform at each wavelength. For simplicity, the central position and FWHM of the echo waveform were unified across all wavelengths, as detailed in Fig. 3. Finally, we simulated the final superimposed waveform at each wavelength, as shown in Fig. 3.

The measured dataset was acquired using the HSL system, consisting of two specific collections. The first collection comprised one stone and one leaf sample, with two varying adjacent distances of 28 and 44 cm, respectively, to depict overlapping levels. The second collection was a scene featuring three targets: two natural targets and a 99% standard reflectance panel, with adjacent distances of 53.00 and 48.85 cm (shown in Fig. 4). To avoid the accidental measurement error, the HSL system recorded the signal three times and the best one was chosen to participate in the decomposition. The design of the second scene aimed to evaluate the component detection capability of the ranking central locations of natural target echoes (Rclonte)-M method in scenarios involving multiple targets, encompassing two natural targets.

The units of central position and FWHM are samples, where two adjacent samples stand for 0.2 ns in this study.

C. Rclonte-M Method

Typically, the natural targets exhibit diverse reflectance properties across different wavelengths [30], leading to distinct intensity values recorded by HSL at various wavelengths. Consequently, some waveform components for natural targets at some wavelengths may not be readily discernible due to weak reflection characteristics or high overlapping levels, making them prone to be stuck in omission or erroneous detection. This Rclonte-M method is designed to decompose the complex return waveform shapes. It is poised to automatically compensate and correct for the missed or false detection of waveform components at the wavelength in question, thereby identifying all the waveform components.

The method can be generally summarized as four steps: waveform preprocessing, parameter initialization, parameter optimization, and reoptimization over calculating the median of parameter sets after ranking central locations of natural target echoes. The workflow of this method is depicted in Fig. 5.

1) Preprocessing: Since noise errors exist in both simulated and measured hyperspectral waveforms, the raw waveforms were first smoothed during decomposition. The Gaussian filtering method was selected for the noise reduction. The FWHM of the Gaussian function was consistent with the transmitted waveform. Gaussian filtering was performed on the return waveform at each wavelength, which facilitates an accurate estimation and optimization in subsequent processes.

For the single-wavelength waveform with only one echo, conventionally, it can be assumed to be a Gaussian distribution function. The raw waveform is often composed of several echoes and is the superposition of these echo signals. Usually, the waveform can be fit and written as the following equation:

$$g(T) = \sum_{i=1}^{n} A_i \exp\left(\frac{-(T - T_i)^2}{F_i^2/(4ln2)}\right) \tag{1}$$

where A_i, T_i, and F_i stand for the amplitude, central location, and FWHM of an echo component within the waveform, respectively, and $g(T)$ stands for the final fit function of the return waveform.

In a specific campaign, the echo intensities and FWHMs are different at various wavelengths for a natural target. In addition, the central locations for the same target are also inconsistent at different wavelengths owing to the inherent time delay effect for HSL, although the central locations are supposed to be the same theoretically [41], [48]. Thus, the following Gaussian mixture model was employed to fit the hyperspectral waveform generated by the HSL system

$$g_{\lambda_k}(T) = \sum_{i=1}^{n} A_{i,\lambda_k} \exp\frac{-\left(T - T_{i,\lambda_k}\right)^2}{F_{i,\lambda_k}^2/(4ln2)} \tag{2}$$

where $g_{\lambda_k}(T)$ is the final fit function of the return waveform at the wavelength of λ_k.

2) Component Parameter Initialization: The initialization strategy determines the accuracy of parameter optimization. This part devised a strict initialization strategy for HSL waveform data. First, the minimum and maximum (peak) points were acquired through the first derivative. A Gaussian component was assumed to exist in the location of the peak point. Thus, Ai and Ti were used to denote the intensity value and center position of the Gaussian component

$$A_i = y_{\text{pp}} \tag{3}$$

$$T_i = T_{\text{pp}} \tag{4}$$

where y_{pp} is the y value of the peak point and T_{pp} refers to the central location of the peak point. Some adjacent components

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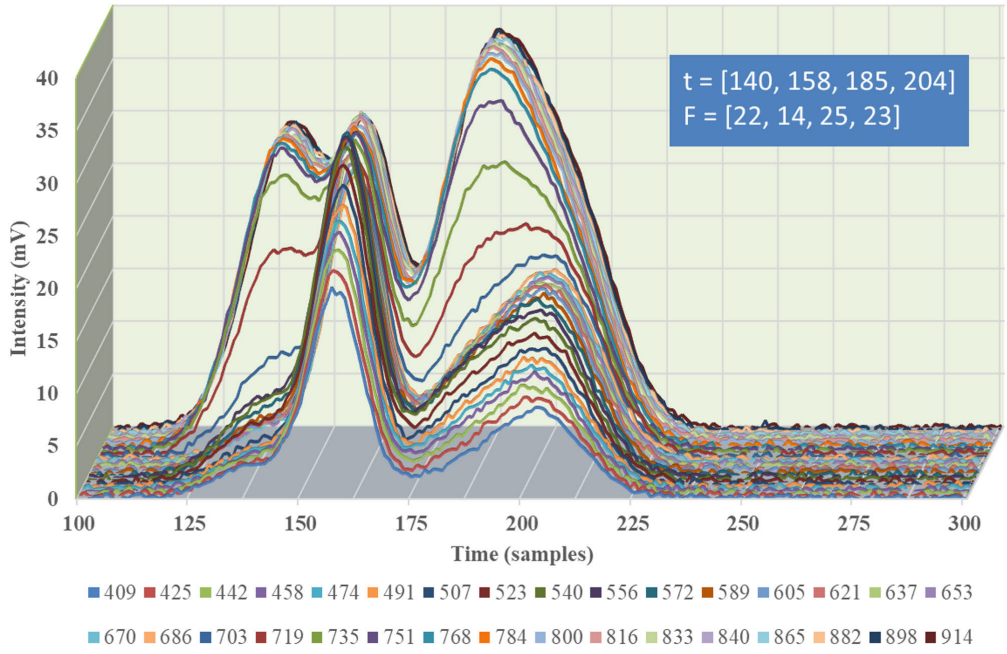


Fig. 3. Simulated dataset for the four natural targets generated from reflectance ratio relationships at various wavelengths.

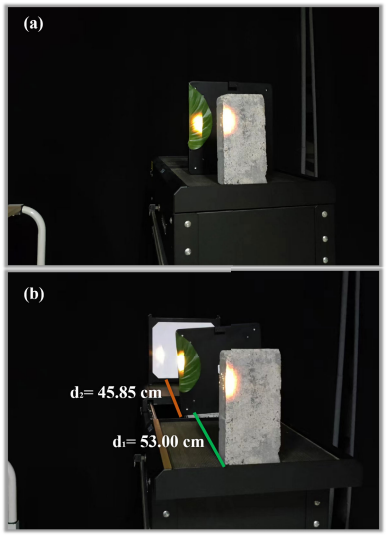


Fig. 4. Photographs of the HSL experiment for the measured dataset.

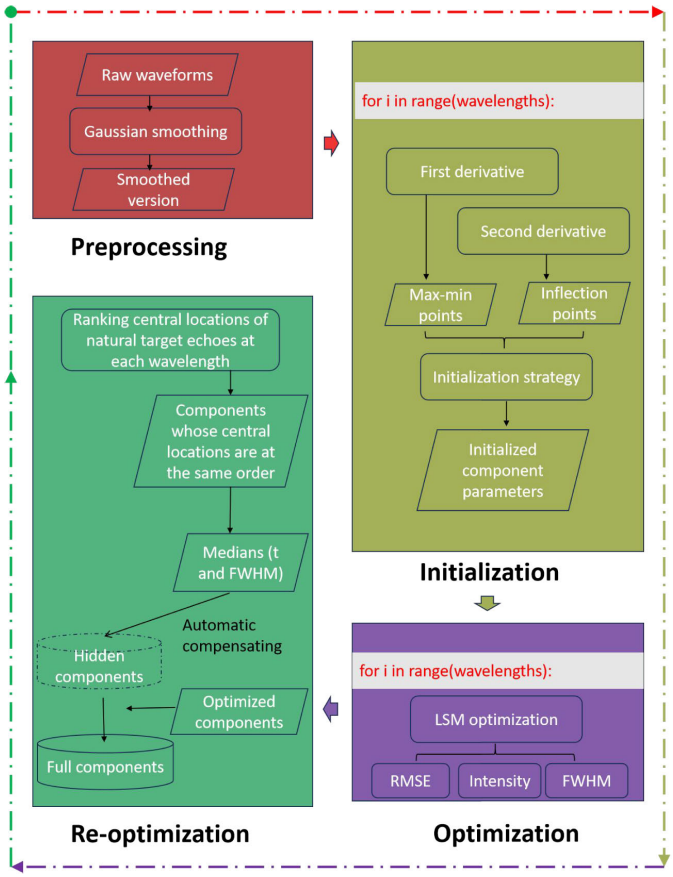


Fig. 5. Workflow of the proposed Rclonte-M decomposition method.

were merged based on the principle that the central position difference of adjacent components should not exceed 1/3 times FWHM of the transmitted laser pulse (F_t represents later). The central location of the newly merged component was expressed as the average of the central locations of two original adjacent components. The y value at the new central location represents the peak intensity of the new component.

Subsequently, the inflection points were identified using the second derivative of the filtered version of the raw waveform. Some of the above-detected components would be deleted in cases where no inflection points are observed on both monotone intervals around a peak point. However, if the inflection points are detected on a single monotone interval or both monotone intervals, the Gaussian component is regarded as valid, and the corresponding FWHM is calculated based on the following principles.

When only one inflection point is observed on the single monotonic interval, the computation of FWHM is

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initialized as follows:

$$F_i = 2\sqrt{2ln2}|l_n - T_i| \tag{5}$$

where l_n stands for the central location of the nearest inflection point in the monotone interval to the peak point.

If the inflection points exist on both monotonic intervals around the peak point, the FWHM can be initialized as follows:

$$F_i = \sqrt{2ln2}|l_{\text{nlo}} - l_{\text{nro}}| \tag{6}$$

where l_{nlo} and l_{nro} stand for the nearest inflection point locations to the peak point in the left and right monotone interval. The above Gaussian components are flagged as "important."

Furthermore, it should be mentioned that a novel component will be generated in cases where multiple inflection points exist in a single monotone interval. The central location and FWHM of this newly added component are expressed as follows:

$$T_i = \left(l_{far1} + l_{\text{far2}}\right)/2 \tag{7}$$

$$F_i = \sqrt{2ln2}|l_{\text{far1}} - l_{\text{far2}}| \tag{8}$$

where l_{far1} and l_{far2} denote the locations of the two farthest inflection points in a single monotone interval.

These newly generated Gaussian components are then flagged as "undetermined."

3) Component Parameter Optimization: The nonnegative least-squares method (LSM) was employed for parameter optimization. Initially, all components identified as "significant" are chosen to partake in the optimization process. If the optimization outcomes fail to meet the criterion of root-mean-square error (RMSE) being less than twice the standard deviation of noise, components labeled as "undetermined" are subsequently involved in the optimization process one by one. Optimization at each wavelength does not end until the RMSE satisfies the criterion of RMSE being less than twice the noise standard deviation.

4) Reoptimization: The reoptimization process initially employs a ranking way for the Gaussian components' central locations within the waveform at each wavelength, such as descending order. Subsequently, a series of components sharing the same central location order across various wavelengths are grouped into a set and sorted once more based on the central locations. For component set with the same central location order, it is considered to correspond to the same natural target with a high probability. Nevertheless, some noisy Gaussian components may also be identified and added. Therefore, the way to calculate the average values for central location and FWHM as the reference parameter sets may bring in the influence of noise and cause inaccuracy in the reference central position, which may lead to local optimum in the LSM optimization process. Along these lines, the median calculation is considered a potential alternative. After the set is sorted, the median values for the central location and FWHM from this ranked set are then obtained as the reference parameter sets to supplement the hidden weak or overlapping Gaussian components. Finally, a thorough examination of parameters, including peak intensity and FWHM, is conducted. Specifically, only the components with peak intensity greater than three times the noise standard deviation and FWHM bigger than that of the transmitted laser pulse are retained.

Method 1 Reoptimization in Rclonte-M Method

Input: optiComSets (first optimized Gaussian components at each wavelength)
Output: reOptiComSets(final re-optimized components)
Do
  for ith in rang(wavelengths)
   rankedComSets[i] = **RankingCenLoc**(optiComSets[i])
  end for
 // totalNumCom to get how many components exist at most
 potentialCom = [len(x) for x in rankedComSets]
 totalNumCom = max(potentialCom)
 // finalLocs to record the final reference central location sets, and final FWHMs to record the final reference FWHM sets
 for jth in range(totalNumCom)
 // waveList is the wavelength set owning (totalNumCom -jth) echoes.
 if len(waveList) >= 1/8∗len(wavelengths)
   cenLocList = CollectCenLoc(waveList)
   FWHMList = CollectFWHM(waveList)
   medianLoc = **Median**(cenLocList)
   medianFWHM = **Median**(FWHMList)
   finalLocs.append(medianLoc)
   finalFWHMs.append(medianFWHM)
   break
 else
   continue
 end for
 for ith in rang(wavelengths)
   Automatic compensating for hidden components using the final parameter sets [finalLocs, finalFWHMs] and Re-optimization using LSM
 end for

D. Accuracy Evaluation

Four indicators are utilized to evaluate the waveform decomposition performance and subsequent tasks. Specifically, they are the coefficient of determination (R^2), RMSE, relative RMSE (rRMSE), and relative neighbor distance error (RNDE)

$$R^2 = 1 - \frac{\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2}{\sum_{i=1}^{n}\left(y_i - \bar{y}\right)^2} \tag{9}$$

$$\text{RMSE} = \sqrt{\frac{1}{\text{n}}\sum_{i=1}^{n}\left(\hat{y}_i - y_i\right)^2} \tag{10}$$

$$\text{rRMSE} = \text{RMSE}/\bar{y} * 100\% \tag{11}$$

where y_i denotes the raw measured waveform data, $\hat{y}_i$ stands for the fit waveform data, and $\bar{y}$ is the average value of the

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raw measured waveform data

$$\text{RNDE} = |d_e - d_t|/d_t \tag{12}$$

where d_e stands for the estimated distance between the two adjacent targets and d_t is the true distance.

III. Results and Discussion

Two datasets were generated to evaluate the decomposition performance of the proposed method, and the outcomes were also compared with the MSWD method. Besides, we explored the ranging accuracy of adjacent components for our HSL system and the spectrum retrieval ability within a footprint along the laser track using the HSL system. Note that in waveform decomposition, the effective echo range encompassing target echo information is retained in decomposed results.

A. Decomposition for Simulated Waveforms

Fig. 6 shows the decomposed results for the simulated data by Rclonte-M and MSWD methods. Overall, the Rclonte-M method completely detected the components corresponding to the four targets in the simulated scene. Notably, the Rclonte-M method successfully identified the weak and overlapping components, such as the first (green leaf) and fourth (sand) target echoes, compared with the MSWD method. Taking the decomposed result at 670 nm as an example, the Rclonte-M method completely detects the four natural targets, while for MSWD, the components representing the green leaf (the first) and sand (the fourth) were missed within the waveform. Apparently, from Fig. 6, the detection results by MSWD are incomplete, and the four components cannot be detected entirely at most wavelengths. This phenomenon can be attributed to the different initialization strategy. On the other hand, the principle of descending order of the intensity also affects the final complementing for the hidden components. For instance, for the MSWD part, the first natural target was detected at some wavelengths of 735–914 nm. Theoretically, those components corresponding to the first natural target could be employed to complement the waveform at which the first natural target was not detected. However, the descending energy order will make these components classified into different sets sharing the same energy order, resulting in the failure to detect the first component at most wavelengths. It demonstrates that MSWD owns the weaknesses in the weak waveform compensation for multiple natural target echo detections.

The average values of each component parameter derived through the Rclonte-M method are presented in Table I. A noteworthy observation is that, compared with the initial input central location and FWHM parameters, there are no significant differences. This observation reveals that Rclonte-M has a full and accurate component detection ability.

Fig. 7 details the distribution curves of R^2, RMSE, and rRMSE for all 32 wavelengths using the two methods. It is clear that the R^2 curve of Rclonte-M is located between 0.95 and 1, above the R^2 curve of the MSWD method. Besides, the RMSE and rRMSE of the MSWD method are apparently larger than that of the proposed Rclonte-M method. The quantitative analysis of the results in Fig. 6 shows that the RMSE average of Rclonte-M is about 0.21 mV, the R^2 average is 0.97, and rRMSE is controlled at an average of about 3%. By contrast, for MSWD, the R^2 average (0.87) is smaller than Rclonte-M and R^2 has a larger variation range (0.78–0.93). Besides, the RMSE and rRMSE average values (1.07 mV and 17%) are both five times larger than those of Rclonte-M. It can be concluded that the fitting performance of Rclonte-M is significantly superior to the MSWD.

TABLE I
Average Rclonte-M-Retrieved Parameter Values of Simulated Waveforms

No.	T_i	F_i
1	140.00	21.95
2	157.99	14.01
3	185.45	25.65
4	204.07	22.88

B. Decomposition for Measured Waveforms

The measured dataset containing two measured distance conditions was also utilized to evaluate the decomposition performance of the proposed method. The wavelengths with a high signal-to-noise ratio (SNR) (491–800 nm) were utilized to participate in the decomposition experiments. As shown in Fig. 8, we show the decomposed results for measured waveforms. For the measured waveforms at distances of 28 and 44 cm, Rclonte-M can entirely detect two natural targets. Thus, the superposition results of the natural target components by Rclonte-M fit the original waveform data points well (the red line). For the MSWD method, when the adjacent distance is 28 cm, only a few wavelengths detect the two components, which is mainly related to the component parameter initialization strategy.

When the adjacent distance is set as 44 cm, the MSWD method detects two components at more wavelengths but not all wavelengths. The energy descending strategy in the reoptimization step ultimately fails to supply reasonable compensation for other wavelengths where only one component is detected. Clearly, the proposed Rclonte-M method achieved better results at various wavelengths.

The box plots of error indicators were also drawn to show the overall fitting performance. As shown in Fig. 9, R^2 of the Rclonte-M method at 28 and 44 cm owns the shorter range with an average value of about 0.83 and 0.84, which are greater than those of 0.57 (28 cm) and 0.65 (44 cm) of MSWD method. Besides, the Rclonte-M method has less box volume and smaller error for indicators of RMSE and rRMSE [shown in Fig. 9(b) and (c)].

Regarding RMSE, the averages of Rclonte-M at 28 and 44 cm are 0.35 and 0.28 mV, respectively, whereas the MSWD method owns bigger values of 1.06 and 0.75 mV. Furthermore, the overall ranges of rRMSE of Rclonte-M method at different distances are lower than those of the MSWD method, with average values of 42% and 34%. It indicates that the proposed method outperforms MSWD and has a higher fitting accuracy for the measured waveform data.

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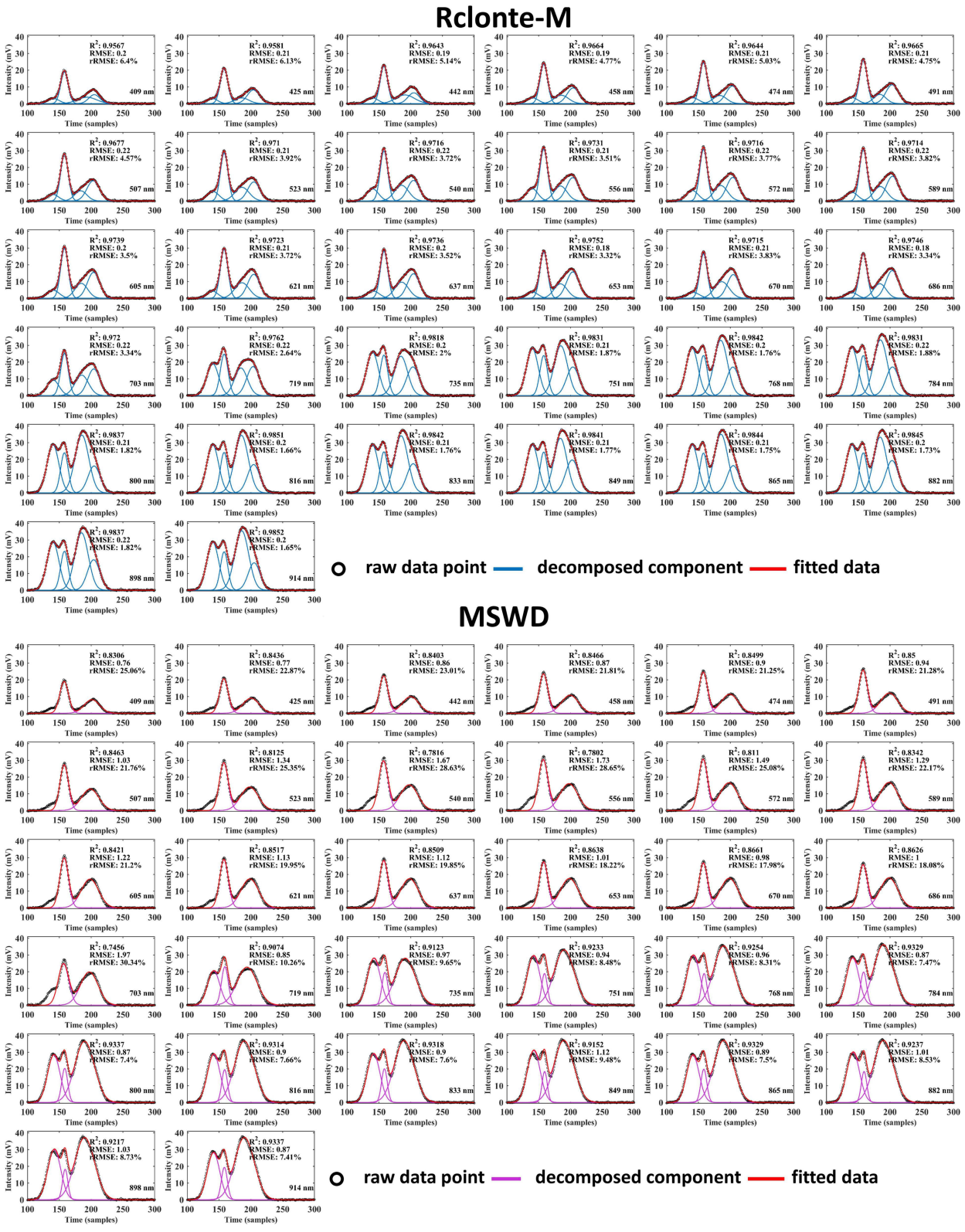


Fig. 6. Decomposed results and fit data of (a) Rclonte-M method and (b) MSWD method for simulated waveforms.

Although the Rclonte-M method has a significant advantage in terms of the component detection ability and the decomposition performance surpasses the MSWD on both simulated dataset and measured dataset with two natural targets, the potential to detect the components of more targets is still to be explored. Therefore, the measured waveform generated from Fig. 4(b) scene with three targets included (two natural targets and a Lambertian target) was utilized to help analyze the decomposition performance of the Rclonte-M method. As shown in Fig. 10, apparently, three components of

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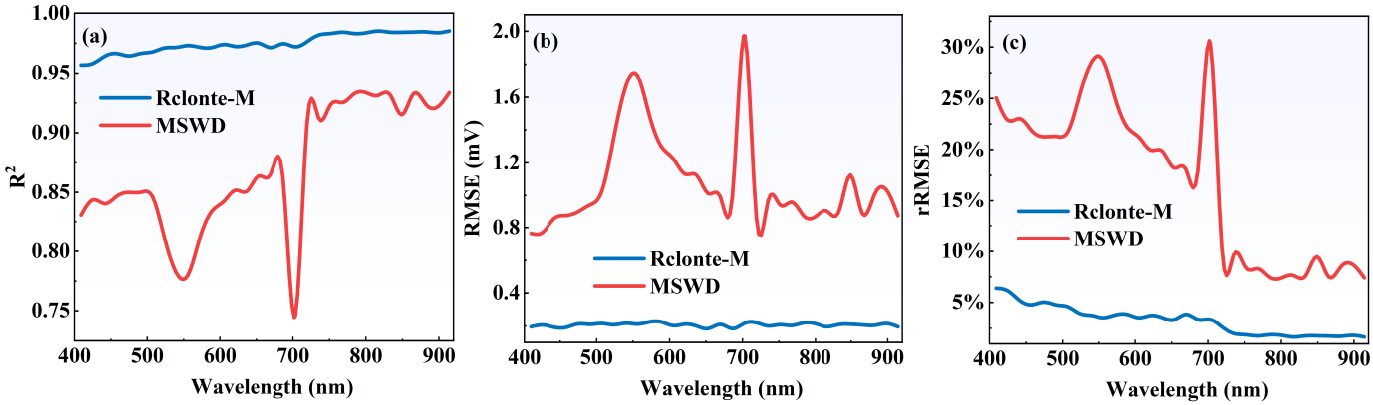


Fig. 7. Fitting accuracy and error statistics of Rclonte-M method and MSWD method for the simulated dataset. (a) R^2, (b) RMSE, and (c) rRMSE.

TABLE II
RETRIEVED DISTANCE VALUES AT VARIOUS WAVELENGTHS USING THE DECOMPOSED RESULTS BY RCLONTE-M METHOD FOR THE FIRST MEASURED EXPERIMENT SCENE

True distance (cm)	Wave. (nm)	T1	T2	Retrieved distance (cm)	True distance (cm)	Wave. (nm)	T1	T2	Retrieved distance (cm)
28	491	304.32	312.88	25.68	44	491	307.59	319.84	36.73
	507	303.81	312.58	26.31		507	304.78	316.93	36.43
	523	303.34	311.24	23.69		523	304.27	317.53	39.75
	540	305.11	314.07	26.87		540	306.33	319.35	39.02
	556	305.21	312.27	21.14		556	306.01	318.83	38.44
	572	303.75	311.84	24.26		572	304.88	317.05	36.49
	589	303.29	310.97	23.03		589	304.49	316.54	36.13
	605	302.27	311.05	26.34		605	303.36	315.12	35.28
	621	302.18	309.46	21.84		621	302.57	313.60	33.06
	637	301.12	308.61	22.45		637	301.74	313.25	34.52
	653	301.12	308.17	21.13		653	301.92	312.44	31.53
	670	303.93	311.85	23.75		670	304.66	316.88	36.63
	686	303.00	311.02	24.04		686	304.47	318.36	41.63
	703	302.23	311.99	29.25		703	303.67	317.97	42.89
	719	301.97	311.52	28.64		719	303.46	316.95	40.42
	735	301.50	311.93	31.27		735	302.83	316.66	41.49
	751	300.56	310.82	30.76		751	301.37	314.49	39.34
	768	301.10	312.30	33.56		768	300.63	314.15	40.54
	784	302.86	314.35	34.44		784	304.36	314.49	30.36
	800	308.24	314.26	18.05		800	305.25	316.97	35.16

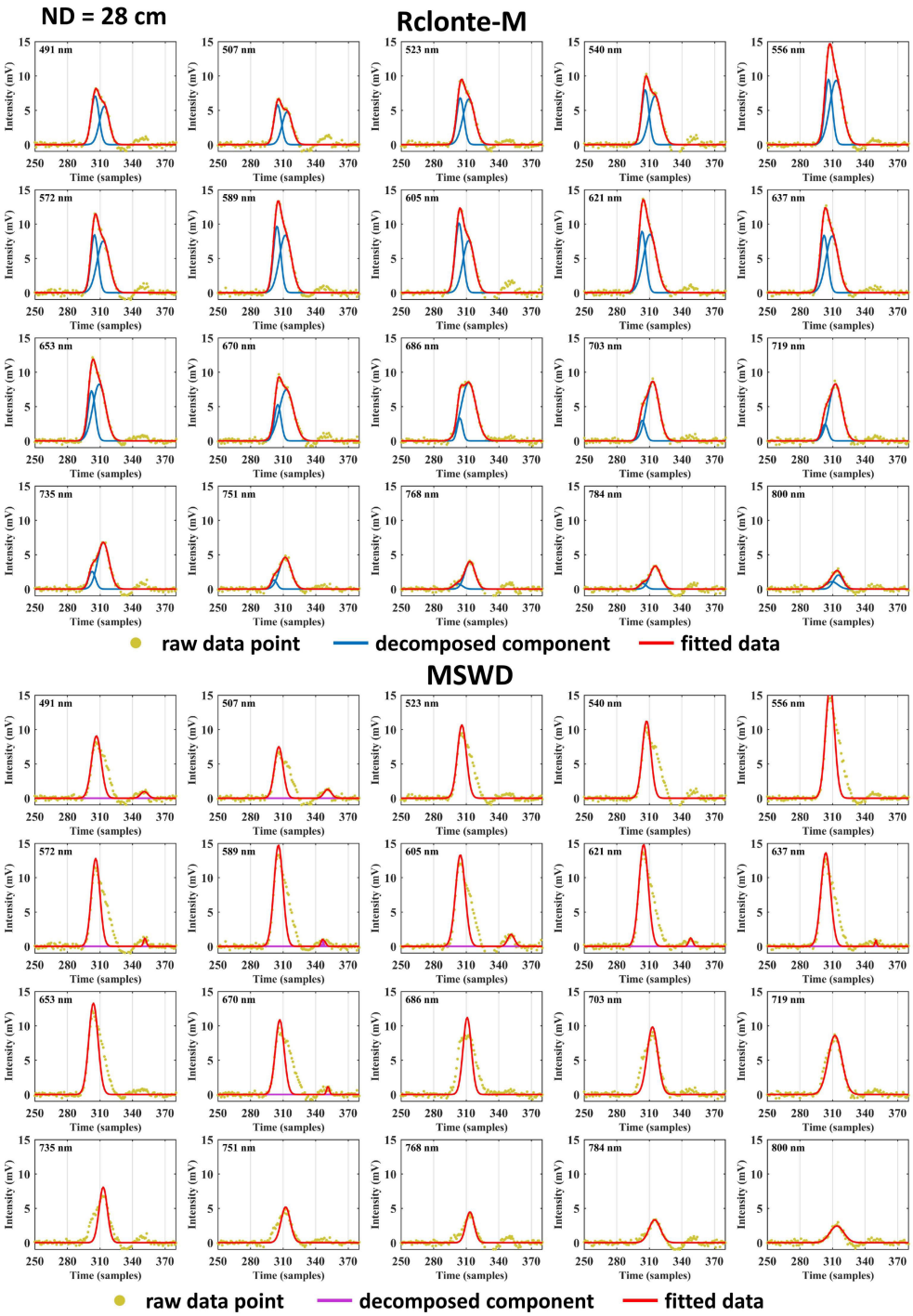
* Ti represents the central location of the related target.

targets, corresponding to stone, leaf, and standard reflectance panel, were detected at most wavelengths, apart from 751 and 800 nm. The fit data also indicate that the decomposed results fit the raw data well. The R^2 average is 0.89, and the RMSE average is 0.25 mV, which proves that the Rclonte-M even has an overall good decomposition performance. Besides, the rRMSE average is 19%, which is smaller than that of 42% and 34% of the measured dataset in Fig. 4(a) at 28 and 44 cm. Nonetheless, it should also be noted that only two components were detected at 751 and 800 nm for this measured scene. It is inferred that although Rclonte-M has a compensation step for the missed component, it was eventually eliminated because the value of FWHM and intensity did not meet the requirement. From the overall view, it can be argued that the Rclonte-M method owns a great component detection ability and holds promise in decomposing more complex waveform data.

C. Ranging Accuracy of Adjacent Components for HSL System

Based on the experiment results, we further evaluate the ranging accuracy of adjacent components for our HSL system. Table II shows the component central locations and corresponding adjacent distance values at different wavelengths for two sets of experimental distances (28 and 44 cm) in measured scene 1. In order to display these ranging results more visually, the results in Table II were plotted visually in

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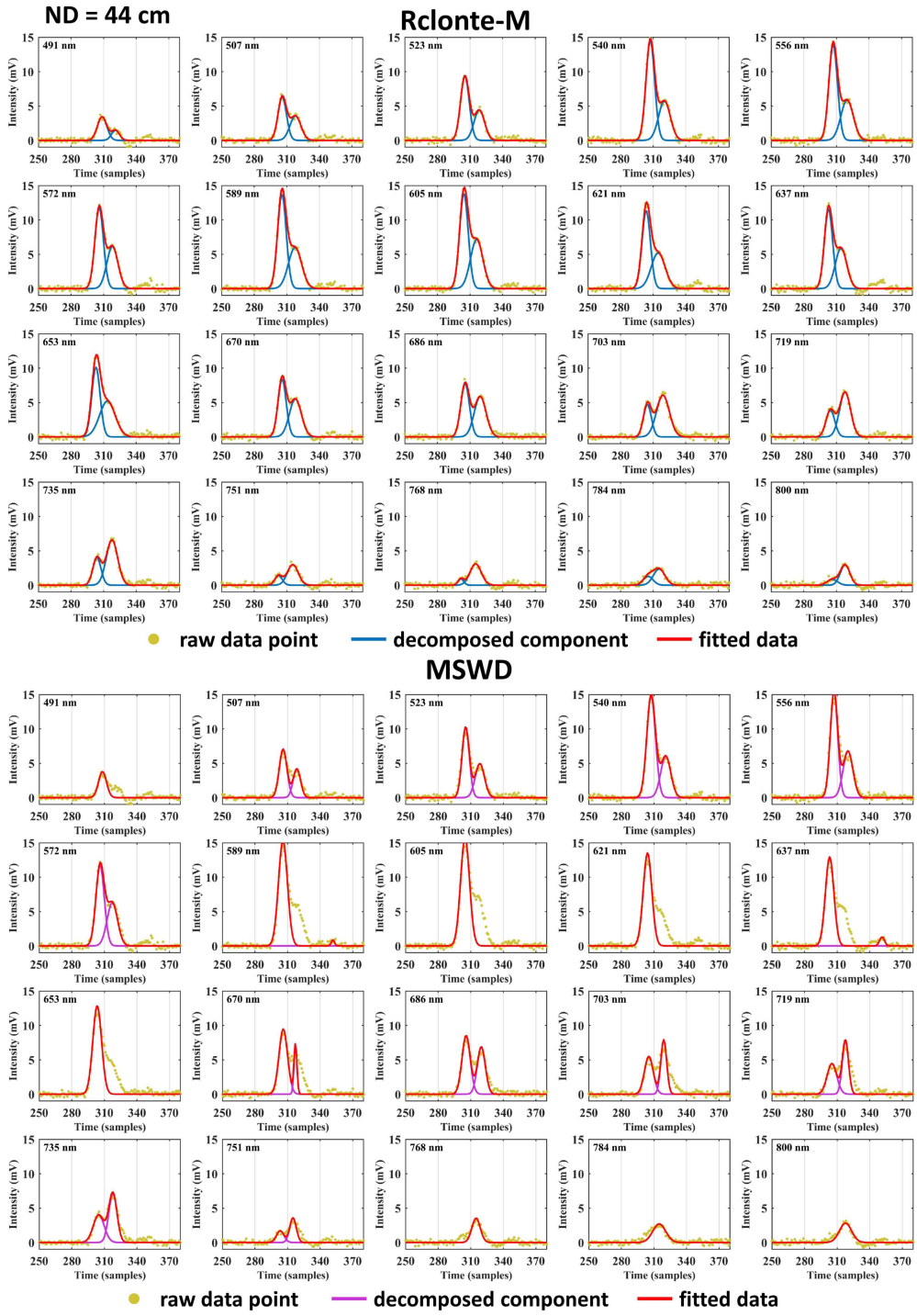
(a)

Fig. 8. Decomposed and fit results for measured waveforms at different distances of (a) 28 cm.

the line chart (see Fig. 11). Clearly, from Fig. 11(a) and (b), the retrieved distances are distributed near the actual distance line of 28 and 44 cm. For the experiment with a distance of 28 cm, the retrieved distance average from the decomposed result by Rclonte-M is 25.83 cm with the standard deviation of 4.37 cm, whereas the distance experiment at 44 cm shows that the retrieved distance average obtained by decomposition results is 37.29 cm with a standard deviation of 3.42 cm. Furthermore, the RNDE curves were also illustrated in Fig. 11(c). The average values of the RNDE of the two sets of experimental ranging errors are 0.149 at 28 cm and 0.152 at 44 cm.

Subsequently, the decomposition results for the measured three-target scenario [see Fig. 4(b)] were also utilized to evaluate the ranging accuracy of adjacent components for our HSL system, and the retrieved distance values were shown in Table III. In addition, the visual distance curve chart and

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(b)

Fig. 8. *(Continued.)* Decomposed and fit results for measured waveforms at different distances of (b) 44 cm by the Rclonte-M method and MSWD method.

RNDE changing trend with wavelength were displayed in Fig. 12. Notably, from Fig. 12(a) and (b), on the whole, the retrieved distance values are distributed near the true value lines, and the error fluctuation occurs at the beginning and end of the wavelength range. It demonstrates that the noise of the middle part of the wavelength is smaller than that of the beginning and end sides, obtaining a better ranging performance. The distance average value of the first preset adjacent distance group is 51.03 cm with a standard deviation of 3.05 cm. Besides, the average value of the second preset adjacent distance is 39.89 cm with a standard deviation of 6.20 cm. The adjacent component ranging results for the three measured collections initially indicate that the HSL system we designed has an overall ranging error of about 7 cm (the biggest standard deviation with the true distance) for adjacent components. Besides, the RNDE average values are 0.038 at 53.00 cm and 0.130 at 45.85 cm. The overall biggest RNDE for the three collections is limited to about 0.160.

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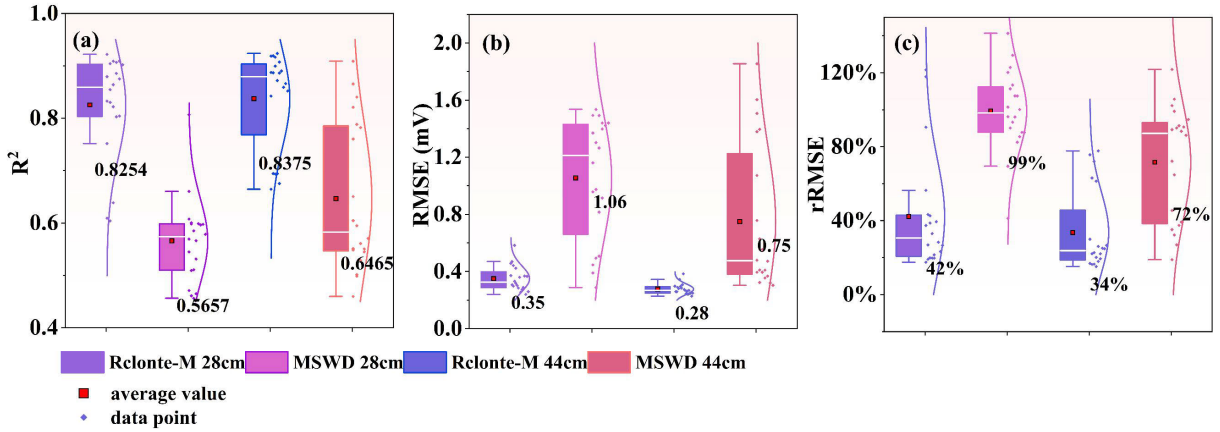


Fig. 9. Boxplot of (a) R^2, (b) RMSE, and (c) rRMSE statistics for measured waveforms by Rclonte-M and MSWD methods.

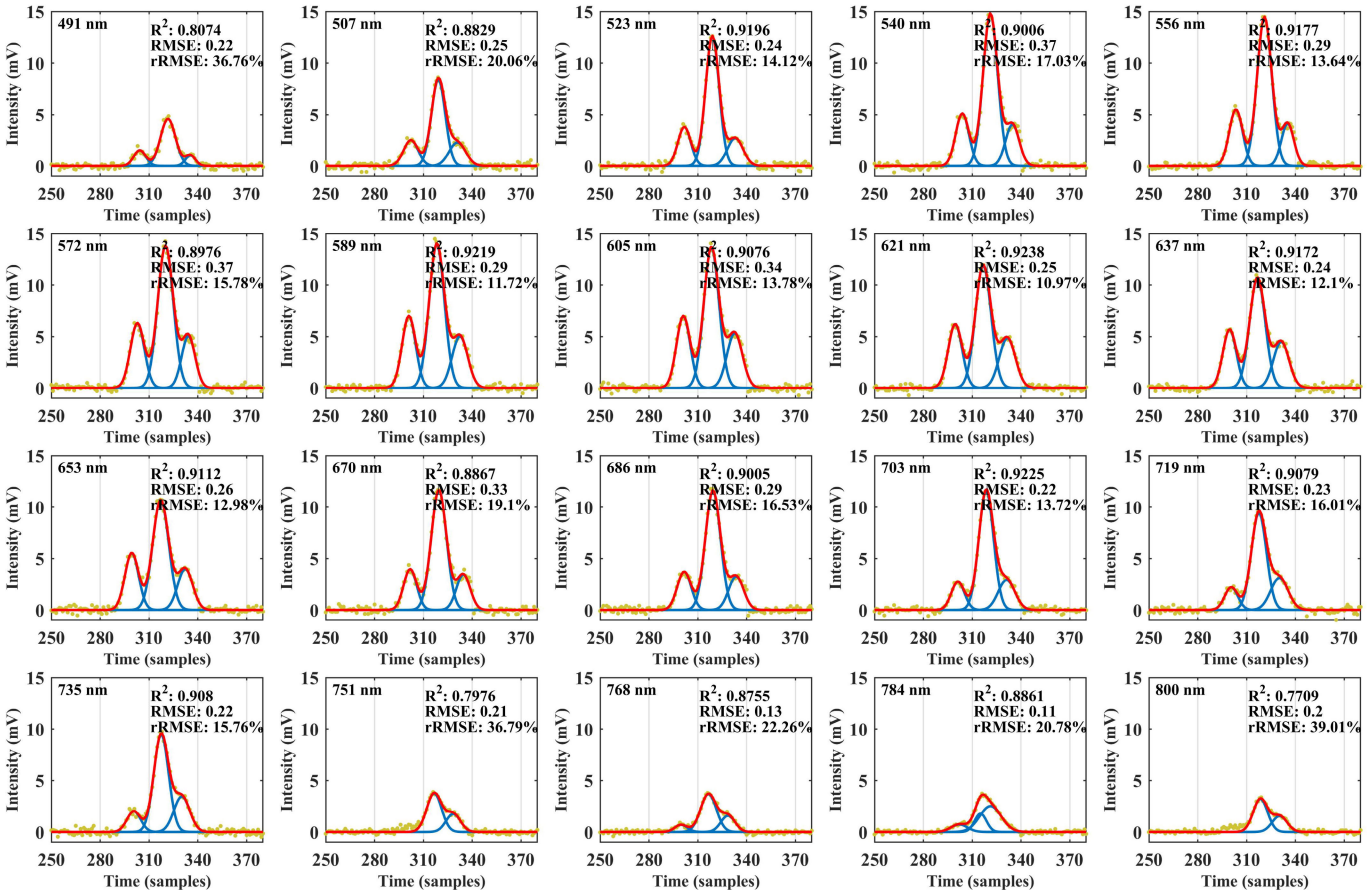


Fig. 10. Decomposed results and fit curves for measured multiple targets' waveforms in Fig. 4(b) by the Rclonte-M method.

D. Ability of HSL to Retrieve Spectra of Natural Targets Along Laser Path

The spectral characteristics were extracted in this study, using the Rclonte-M that has been proven to perform well from the above analysis, to demonstrate the spectrum retrieval ability of HSL, for scenes with multiple natural targets within the same footprint along the laser path.

First, the spectral intensity with wavelength for the simulated waveform data was collected using the decomposition results and shown in Fig. 13. It reveals that the variation trends of spectral intensity are basically consistent with the changing of the reflectance curves in Fig. 2. Four intensity spectra were accurately extracted by the simulated HSL signals. For the measured data, taking the third measured collection with three targets as an example, we normalized the component intensity of the natural targets (stone and leaf) using the intensity of a standard reflectance panel and obtained the normalized reflectance, as shown in Fig. 14. The results basically reflect the ability of the current HSL system prototype to invert the complex reflection characteristics of multiple targets

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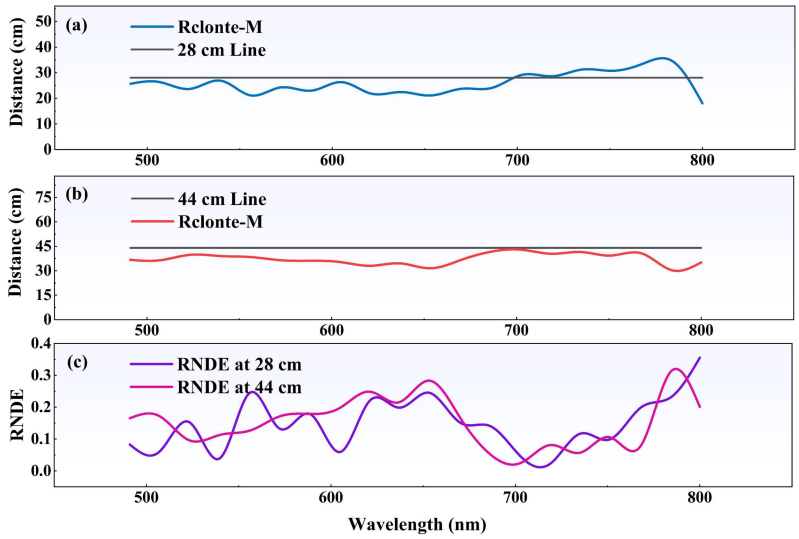


Fig. 11. Retrieved distance curves with the wavelength of HSL for the first measured experiment scene at (a) 28 cm, (b) 44 cm, and (c) RNDE curves using the Rclonte-M method.

TABLE III

RETRIEVED DISTANCE VALUES AT VARIOUS WAVELENGTHS USING THE DECOMPOSED RESULTS BY THE RCLONTE-M METHOD FOR THE SECOND MEASURED EXPERIMENT SCENE

Wave. (nm)	T1	T2	T3	Retrieved distance 1 (cm)	Retrieved distance 2 (cm)
491	303.30	320.66	334.40	52.05	41.21
507	301.30	318.16	329.86	50.54	35.07
523	300.72	317.93	331.97	51.58	42.10
540	302.90	320.15	333.68	51.69	40.56
556	302.42	319.96	334.19	52.58	42.67
572	301.84	319.12	333.18	51.81	42.15
589	300.06	317.21	331.43	51.44	42.63
605	300.10	317.35	331.65	51.72	42.86
621	298.53	315.75	330.58	51.64	44.47
637	298.54	315.62	330.24	51.21	43.83
653	298.41	316.17	331.23	53.25	45.13
670	300.84	318.48	333.48	52.88	44.96
686	300.63	318.27	332.54	52.90	42.77
703	300.20	317.58	330.10	52.09	37.56
719	299.56	316.52	329.15	50.85	37.86
735	299.66	316.49	329.14	50.47	37.92
768	298.68	315.52	327.52	50.51	35.96
784	300.99	314.09	320.20	39.27	18.29

* Preset adjacent distance for 'Retrieved distance 1' is 53.00 cm; Preset adjacent distance for 'Retrieved distance 2' is 45.85 cm

along the laser path within a footprint. Apparently, from the changing trends in Fig. 14(a) and (b), the retrieved spectral measurements can be utilized for stone and leaf identification. Some features of the vegetation spectrum are clearly visible, such as the passive optical absorption valleys in the blue and red wavelengths, the reflection peaks in green wavelengths, as well as the not very steep red edge due to the HSL system itself. The shape of the red edge is close to that of the reflectance curve obtained by HSL when measuring a single blade [50].

E. Discussion

The single-wavelength decomposition method typically falls short in capturing targets' spectral profiles along the laser's path, especially for targets with low reflectivity at some wavelengths or targets with too adjacent distance. The method proposed in this study established a strict parameter initialization strategy, which guarantees a comprehensive preliminary detection of components. In the reoptimization step, the median values for the central location and FWHM from the central location ranked set are then selected as the reference

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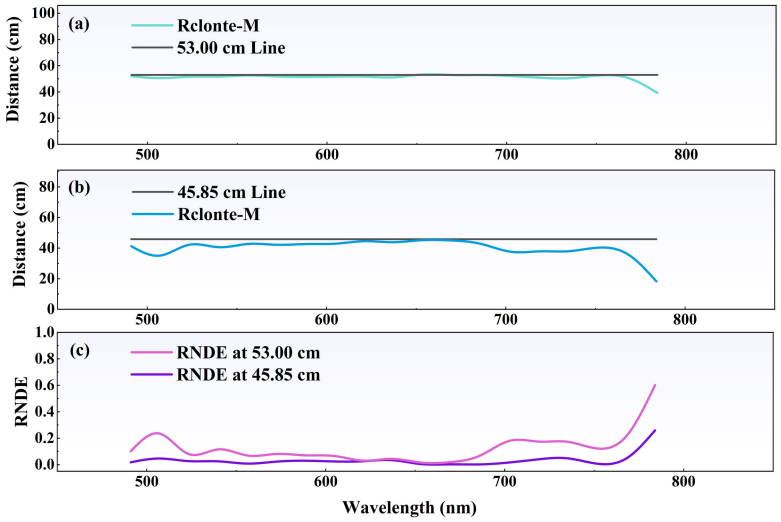


Fig. 12. Retrieved distance curves with wavelength of HSL for (a) and (b) second measured experiment scene and (c) RNDE curves.

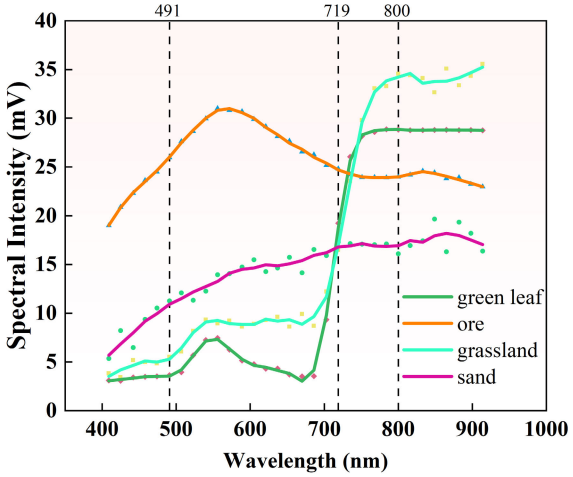


Fig. 13. Retrieved spectral results with wavelength for the simulated waveform data.

parameter sets to complement the hidden weak or overlapping components for the natural targets. The proposed Rclonte-M method enriches the algorithm library for HSL waveform data processing. Compared with the MSWD method [40], the component detection capability is greatly improved and can be applied to scenes with multiple natural targets, not only Lambertian targets.

About the Rclonte-M, the compensation ability for weak echoes after taking the medians is worth being emphasized again. Taking the waveform at 572 nm in the first measured scene with an adjacent distance of 28 cm as an example, the results before and after the reoptimization are shown in Fig. 15. Before the second optimization, only one strong echo was detected at 572 nm, and the second hidden component in the waveform was not detected. However, the medians, namely, the reference parameter sets collected after the first optimization by us (listed in Table IV), were then added to the reoptimization step for the update. The reference parameter sets complemented the component deficiency. Finally, the two

TABLE IV
REFERENCE PARAMETER SETS FOR THE FIRST MEASURED DATASET AT 28 CM

No.	T_i	F_i
1	305.21	9.64
2	312.30	10.98

components were successfully detected, and the fit result had a good consistency with the original data. It should also be emphasized that the kernel function in the method could also be replaced by other potential functions according to the actual situation. Besides, the required valid criteria for the final components, such as the peak intensity greater than three times the noise standard deviation and units of central location and FWHM, can be adjusted and flexible in actual tasks. In addition, some detailed parts of decomposition process can also be adopted for the conventional LiDAR tasks.

Compared with the published methods in HSL waveform processing in the literature [40], [41], this study utilized the scenarios with multiple natural targets instead of nonnatural targets for evaluating the decomposition performance of the proposed method and accessed the ranging and spectrum retrieval abilities for the HSL system. In contrast to selecting the Lambertian, artificial, and other plastic targets for demonstration, the scenes with several natural targets are more convincing. Accurate waveform processing technique is vital for radiative effect analysis and correction, such as distance effect [42] and incidence angle effect [5].

Nonetheless, it should be acknowledged that current hyperspectral laser equipment has certain limitations, such as the huge size and weight that cannot easily support large-scale aerial missions. In terms of the HSL, developing more waveform processing methods is necessary, but the increasing improvements in component detection capabilities, ranging capabilities, and spectral inversion capabilities rely much more on developing advanced equipment.

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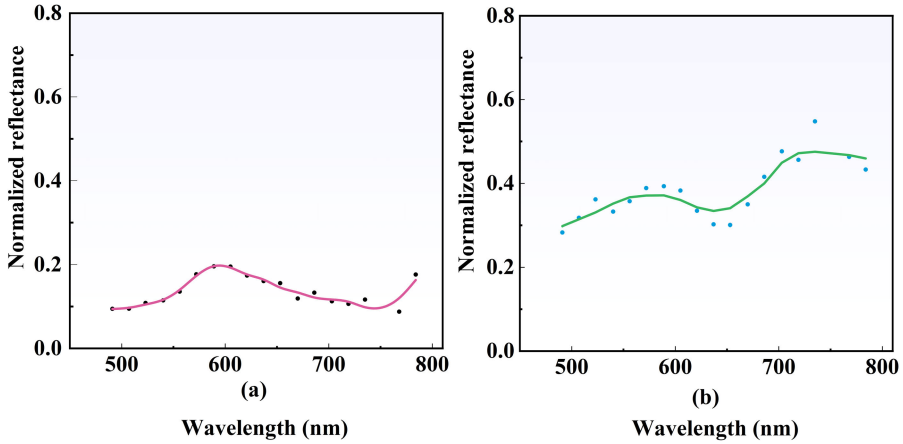


Fig. 14. Retrieved spectral results with wavelength for the second measured three-target included scene in Fig. 4(b). (a) Stone and (b) leaf.

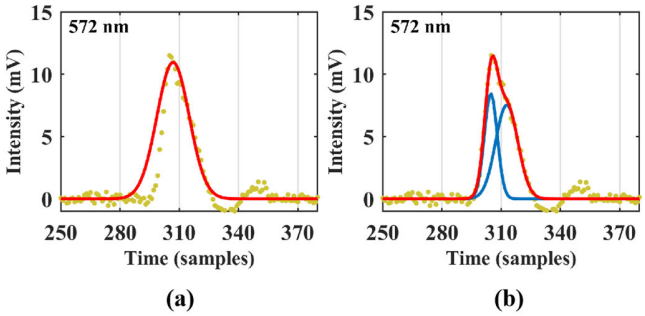


Fig. 15. First (a) optimized result and (b) reoptimized result using Rclonte-M at 572 nm for the first measured dataset at 28 cm.

IV. Conclusion

This study designed a novel HSL waveform decomposition framework and verified it on simulated and measured datasets. Scenarios containing multiple natural targets were constructed in both the simulated and measured datasets, which is conducive to fully examining the practicability of the proposed Rclonte-M method. Component detection, ranging error analysis, as well as spectrum extraction were all carried out. The method proposed in this study can enrich the algorithm library for waveform processing and provide a reference for other multispectral or full-waveform HSL data processing. The main contributions can be summarized as follows.

1) A Rclonte-M method framework was proposed in this study for detecting hyperspectral echo components. The decomposed results of the simulated dataset and the measured dataset both indicate that the method in this study can detect echo components more accurately compared with the MSWD method.

2) The ranging error of adjacent components for the HSL system using decomposition results shows that the current instrument has a ranging error of 7 cm with RNDE limited to 0.160. The Rclonte-M method can improve the range resolution of the HSL.

3) The spectrum retrieval results reveal that the HSL can easily discern multiple targets within the footprint along the laser track. Meanwhile, note that the reflectance extraction accuracy is not only dependent on the decomposition method but also determined by the advanced experiment equipment.

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